Impulse changes momentum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect force over time to momentum change.

SUCCESS CRITERIA - I can...

1. Calculate impulse from force and duration.
2. Match impulse direction to momentum change.
3. Use impulse to find a final velocity.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “impulse” and “average” to explain your model or result. Sentence starters are optional.

impulse: net force acting over time; change in momentum

average: one value representing an interval or a set of values

momentum: mass times velocity, including direction

01 Bring back an earlier idea

An ideal circular orbit has GM=225 m^3/s^2 and r=9 m.
Find speed, inward acceleration and period; explain how velocity changes.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $J = F_{avg}\,\Delta t = \Delta p = p_f - p_i$.
J is impulse; F_{avg} is average net force. Signed force-time area gives impulse; i = initial, f = final.

Worked reasoning: A constant 6 N force acts right for 2 s. Impulse is 12 N·s right, so momentum increases by 12 kg·m/s.

Your turn: Draw the force-time rectangle and shade its area.
Explain why the two impulse units are equivalent.

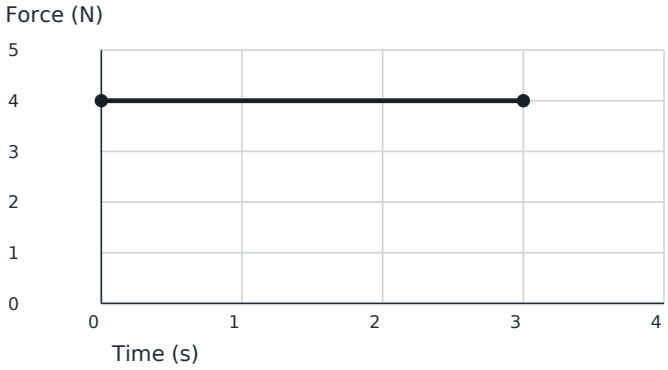
Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

03 Represent and interpret

Read the rectangular force-time graph.
Find the area and state what it means.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

04 Solve and justify

A .50 kg ball changes velocity from +8 to -6 m/s in .020 s.
Find impulse and average net force.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Analyze the reasoning

Two force pulses have equal signed area but different peak heights. What must be equal? What need not be equal?

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Choose a method

At what altitude h/R is a spherical planet's field 1/16 of its surface value? Repeat for 1/36.

Word help: altitude = height above a reference surface

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

07 Exit check

A triangular force pulse has base .20 s and peak 30 N. **Find** impulse and velocity change for a .60 kg object.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 2 kg cart changes velocity from +3 to -2 m/s in 0.1 s. **Find** impulse and average force. **Sketch** a force-time rectangle with the correct signed area.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: 1 2 3